

Trajectories of Charged Particles Trapped in Earth's Magnetic Field

Abstract

This article presents the theory of the motion of **relativistic charged particles** within the Earth's magnetosphere at a level suitable for undergraduate physics curricula. The discussion encompasses individual particle dynamics and **guiding center motion**, deriving the **three adiabatic invariants** associated with the three periodic motions in a dipole field. Twelve computational exercises are provided for use as classroom assignments or independent study. Two of the exercises cover **drift shell bifurcation** and **Speiser orbits**.

1 Introduction

Ions and electrons trapped in the Earth's magnetic field can significantly impact our technology and daily lives. Energetic plasma particles can damage satellite electronics and pose health risks to astronauts. Phenomena such as **auroras** are generated by particles entering the terrestrial atmosphere at polar regions. Conversely, aircraft crews and frequent flyers may accumulate significant radiation doses due to these same particles. These effects intensify during **solar maximum** periods. Occasional extreme solar events can induce ionospheric currents, which translate into significant currents in power lines, potentially causing blackouts. Such events may also disrupt radio communications and GPS signals.

Consequently, understanding and predicting processes within the Earth's **magnetosphere** is of great importance. This paper aims to present the fundamental physics of charged particle motion in a dipolar magnetic field, suitable for students and instructors. The emphasis is placed on numerical computation and the characterization of trajectories through three distinct periodic motions –**cyclotron**, **bounce**, and **drift**– and their associated adiabatic invariants. For a more comprehensive discussion, advanced plasma physics texts should be consulted.

Figure 1 illustrates the primary features of the magnetosphere, the spatial region where the Earth's magnetic field is dominant. Trapped charged particles form the **Van Allen radiation belts**, the **plasmasphere**, and current systems such as the **ring current**, the **magnetotail current**, and field-aligned currents. The Earth's radius ($R_e \approx 6378$ km) serves as a natural length scale. Near the Earth (up to $3\text{--}4 R_e$), the field is approximately dipolar. However, at greater distances, the **solar wind** –a plasma stream carrying solar magnetic fields– induces significant deviations.

Upon encountering the magnetosphere, the solar wind and terrestrial systems do not mix due to the "**frozen-in flux**" condition, which dictates that plasma particles are tied to magnetic field lines except in specific regions like the poles. The solar wind influences

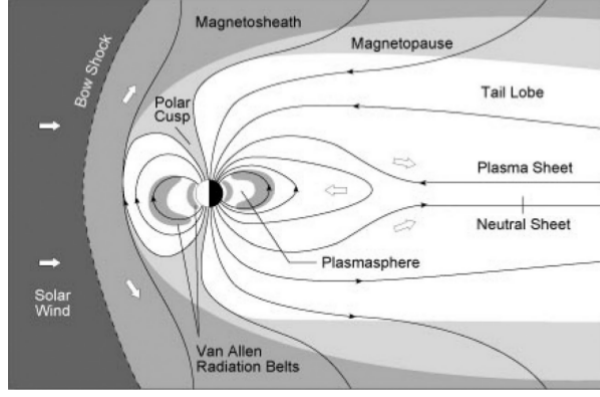


Figure 1: A schematic view of Earth's magnetosphere. The solar wind is coming from the left. (Reproduced with permission from Kenneth R. Lang, Cambridge Guide to the Solar System, 2nd edition. Copyright 2011 by Cambridge University Press.)

the magnetosphere via mechanical and magnetic pressure, compressing it on the dayside. This compression is more pronounced during high solar activity. On the nightside, the field extends over a great distance, forming the **magnetotail**. The Van Allen radiation belts constitute a distinct plasma population characterized by high energy and proximity to Earth, extending from 1000 km to several R_e . These belts consist of electrons with energies of several **MeV** and protons with several hundred **MeV**. The dynamics of these particles are the primary focus of this paper.

2 Particle Trajectory in a Dipole Magnetic Field

The motion of a particle with charge q and mass m in electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields is governed by the **Newton-Lorentz equation**:

$$\frac{d(\gamma m \mathbf{v})}{dt} = q\mathbf{E}(\mathbf{r}) + q\mathbf{v} \times \mathbf{B}(\mathbf{r}) \quad (1)$$

where $\gamma = (1 - v^2/c^2)^{-1/2}$ is the relativistic factor.

Assuming $\mathbf{E} = 0$, the acceleration is always perpendicular to the velocity; thus, the particle's speed (and γ) remains constant. In a uniform magnetic field, particles move in helices parallel to the field vector. The circular component is known as **cyclotron motion** or **gyromotion**, with the **cyclotron frequency** (Ω) and **cyclotron radius** (ρ) defined as:

$$\Omega = \frac{qB}{\gamma m} \quad (2)$$

$$\rho = \frac{\gamma m v_{\perp}}{qB} \quad (3)$$

where $v_{\perp}$ is the velocity component perpendicular to the field. If an electric field exists, it can be decomposed into parallel ($\mathbf{E}_{\parallel}$) and perpendicular ($\mathbf{E}_{\perp}$) components. A non-zero $\mathbf{E}_{\parallel}$ would rapidly accelerate particles out of the region; therefore, the existence of trapped plasma implies $\mathbf{E}_{\parallel}$ is negligible. The perpendicular component induces an **$\mathbf{E} \times \mathbf{B}$ drift**:

$$\mathbf{v}_E = \frac{\mathbf{E}_{\perp} \times \mathbf{B}}{B^2} \quad (4)$$

which is independent of the particle's mass and charge. In this paper, $\mathbf{E}$ is set to zero to simplify the discussion of the guiding center and adiabatic invariants, as magnetic forces dominate the radiation belt dynamics.

The magnetic field of a dipole with moment $\mathbf{M}$ is given by:

$$\mathbf{B}_{dip}(\mathbf{r}) = \frac{\mu_0}{4\pi r^3} [3(\mathbf{M} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{M}] \quad (5)$$

For Earth, $\mathbf{M} = -M\hat{\mathbf{z}}$. In Cartesian coordinates, the field is expressed as:

$$\mathbf{B}_{dip} = -\frac{B_0 R_e^3}{r^5} [3xz\hat{\mathbf{x}} + 3yz\hat{\mathbf{y}} + (2z^2 - x^2 - y^2)\hat{\mathbf{z}}] \quad (6)$$

where $B_0 \approx 3.07 \times 10^{-5}$ T and $x = R_e, y = 0, z = 0$ at the magnetic equator.

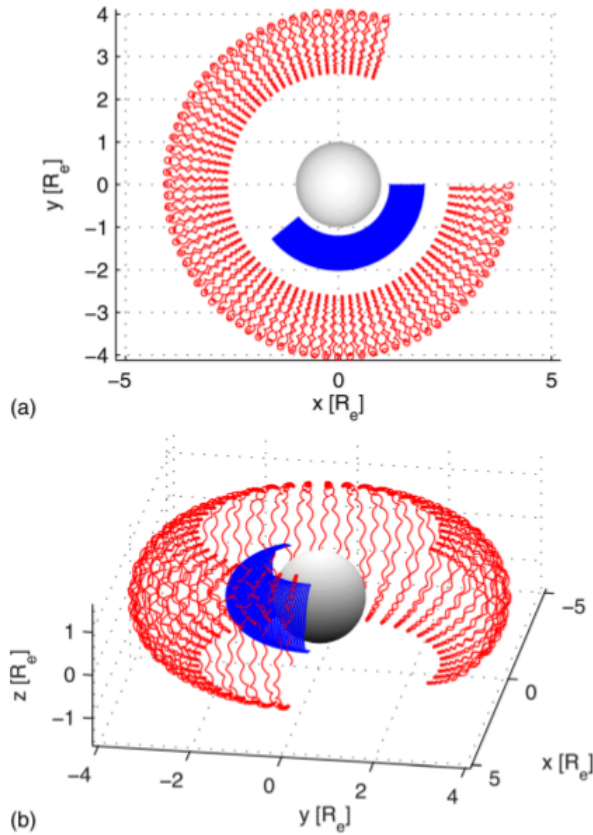


Figure 2: Trajectories of two 10M eV protons in the Earth's dipole field. The dipole moment is in the $-\hat{\mathbf{z}}$ direction. Both figures show the same trajectories from different views.

Figure 2 depicts trajectories for 10 MeV protons. The non-uniformity of the field introduces **bounce** and **drift** motions. **Bounce motion** is the periodic north-south oscillation along field lines, where particles reflect at **mirror points** due to the converging field lines. **Drift motion** involves the azimuthal movement of the particle across field lines. For a proton at $2 R_e$, typical periods are: cyclotron ≈ 0.02 s, bounce ≈ 1.7 s, and drift ≈ 332 s. In a dipole field, the periods are approximated as:

$$\tau_b \approx 0.117 \left(\frac{R_0}{R_e} \right) \frac{c}{v} [1 - 0.4635(\sin \alpha_{eq})^{3/4}] \quad (7)$$

$$\tau_d \approx \frac{2\pi q B_0 R_e^3}{m v^2} \frac{1}{R_0} \left[1 - \frac{1}{3} (\sin \alpha_{eq})^{0.62} \right] \quad (8)$$

where R_0 is the equatorial distance and α_{eq} is the equatorial **pitch angle**.

3 First Adiabatic Invariant

When system parameters vary slowly relative to oscillation frequencies, the system possesses an **adiabatic invariant**. In Hamiltonian mechanics, this corresponds to the action:

$$J = \oint p dq \quad (9)$$

For a charged particle, the canonical momentum is $\gamma m \mathbf{v} + q \mathbf{A}$. The **first adiabatic invariant** (J_1) is obtained by integrating over a cyclotron orbit:

$$J_1 = \oint (\gamma m \mathbf{v} + q \mathbf{A}) \cdot d\mathbf{l} \quad (10)$$

Using Stokes' theorem and assuming a nearly uniform field over the gyro-radius, we derive the **magnetic moment** (μ):

$$\mu = \frac{\gamma^2 m v_\perp^2}{2B} \quad (11)$$

which remains invariant. As B increases, $v_\perp$ must increase to conserve μ ; when $v_\perp$ reaches the total velocity v , the particle reflects at the mirror point $B_m = \gamma^2 m v^2 / 2\mu$.

4 Guiding Center Equations

The **guiding center** is the geometric center of the gyromotion. In non-uniform fields, it experiences a lateral drift. The guiding center approximation is valid when $\rho \ll B/|\nabla B|$. By averaging the motion over a cyclotron cycle, the guiding center acceleration is:

$$\ddot{\mathbf{R}} = \frac{q}{\gamma m} \dot{\mathbf{R}} \times \mathbf{B}(\mathbf{R}) - \frac{q \rho^2 \Omega}{2 \gamma m} \nabla B(\mathbf{R}) \quad (12)$$

The parallel motion is governed by:

$$\frac{dv_\parallel}{dt} = -\frac{\partial}{\partial s} \left(\frac{\mu B(s)}{\gamma^2 m} \right) \quad (13)$$

while the perpendicular drift velocity is:

$$\frac{d\mathbf{R}_\perp}{dt} = \frac{\gamma m}{2qB^2} (v^2 + v_\parallel^2) \hat{\mathbf{b}} \times \nabla B \quad (14)$$

This total drift is the sum of **gradient drift** and **curvature drift**. Because ions and electrons drift in opposite directions, this generates the **ring current**.

5 Second and Third Adiabatic Invariants

The **second adiabatic invariant** (J_2) is associated with bounce motion:

$$J_2 = \oint \gamma m v_{\parallel} ds = 2\gamma m v I \quad (15)$$

where $I = \int_{s_{m1}}^{s_{m2}} \sqrt{1 - B(s)/B_m} ds$. I depends only on the magnetic field geometry and is used to calculate drift paths.

The **third adiabatic invariant** (J_3) is associated with drift motion around the Earth. It is proportional to the **magnetic flux** (Φ) enclosed by the drift shell:

$$J_3 \approx q\Phi \quad (16)$$

These invariants are essential for predicting long-term particle behavior, as purely numerical solutions accumulate errors over time. Drift shells can be constructed geometrically using these invariants, which also serve as dynamic variables for studying particle diffusion.

Note: The document is based on Kaan (2011).

References

- [1] M. Kaan Öztürk Trajectories of charged particles trapped in Earth's magnetic field, Yedite University, Turquia, 2011.